
Learning to Manipulate a New Object from Action-Free Videos

Group 9
McGill University
ECSE 552 — Project proposal

Background

Application. We focus on teaching a robot arm that already manipulates one object to handle a new one from videos alone, without recording new motor commands. This matters for warehouse and home robots, where new items arrive constantly and teleoperating each one is impractical. We study this in simulation with OCBench (UR5e arm, Robotiq gripper) [1].

Problem and need. Visuomotor policies learn by imitating images paired with the motor commands that produced them. Recording these commands requires teleoperation, which is slow and costly: collecting 76k demonstrations took 50 people a year [2]. Videos without commands are cheap (e.g., videos of the same robot generated by video world models [3]), and inverse dynamics models (IDMs) let a robot learn from them: trained on labeled demonstrations, an IDM infers the command between two consecutive frames and labels videos for imitation [4, 5], needing few labeled examples [6]. Our application adds one difficulty: the videos show an object absent from the labeled demonstrations, so the IDM must label images that differ from its training data. Formally, given labeled cube transitions $\mathcal{D}_A = \{(o_t, o_{t+1}, a_t)\}$ and unlabeled block videos $\mathcal{D}_B = \{o_t\}$, we seek an image-only policy $\pi(a | o)$ that succeeds on the block under R1–R6, which requires labeling $\mathcal{D}_B$ accurately without block labels.

Constraints. The design must satisfy the requirements below (red values to be agreed).

ID	Requirement	Definition	Target
R1	Supervision	Block data are RGB frames only, without actions or states	0 block labels
R2	Label budget	Labeled cube demonstrations, identical for every method	300 demos
R3	Inputs	Policy sees only images; no simulator state in any model	Images only
R4	Target	Block success, as share of the gap from cube-only to oracle	$\geq 50\%$ of gap
R5	Retention	Drop in cube success after adding block data	≤ 10 points
R6	Real time	Policy inference per step at 10 Hz (offline IDM exempt)	< 100 ms

Aims/Goals

Goals. Train an image-only policy for the block from cube demonstrations and block videos alone. R4 and R5 are the performance targets; R1–R3 and R6 hold by design.

Scope. We use the visual single-block task of OCBench. The source object is its cube; the target is a new elongated block, with adapted expert grasps. Cameras, lighting, goal, and initial positions stay fixed, so the object is the only change. Human videos, real hardware, other shifts, and large pretrained models are out of scope.

Proposed solution. We adapt the IDM to the block before labeling. Since the arm looks the same in both domains and only the object changes, the IDM receives each frame with its difference to the next, emphasizing arm motion, and domain-adversarial training [7] makes the remaining features

indistinguishable between cube and block while they still predict cube commands. The adapted IDM labels the block videos for policy training; we also compare aligning the policy instead.

Methodology

Design and implementation. [TO WRITE: A5–A6, about 110 words. Architecture of the proposed solution (inputs, outputs, losses), how it is trained, and the tools used.]

Validation, testing, and evaluation. [TO WRITE: A7, about 80 words. Data splits, model selection without block labels, metrics tied to R1–R6, seeds and episodes (agreed: 3 seeds $\times$ 100 evaluation episodes per method and object, from initial states shared by all methods, fixed before seeing results).]

Alternatives. [TO WRITE: A8, about 100 words. Controls and baselines run on our data, and why the proposed solution could be superior.]

Feasibility and resources

Dataset, resources, and plan. We generate all data in OCBench on CPU MuJoCo: 300 labeled cube demonstrations and 200 block videos, with true block actions stored only for evaluation; no licensing or ethics constraints apply. Training needs one GPU [TEAM: source to confirm], about 4 GPU-hours per run and 100 in total. Open-source domain-adaptation and imitation code lowers risk. A pilot by October 21 checks the expert, the cube policy’s drop on the block, and the growth of IDM error. Core methods follow by November 1; evaluation by November 18.

Risk analysis

Pitfalls and mitigation. The shift may be too small to matter, as robot and cameras are unchanged; the pilot measures it, and we can enlarge the change. The block may need actions absent from cube demonstrations; invariant features can then discard information needed to predict them [8], which per-component errors will expose. Closed-loop failure is mitigated by chunking; expert failure on the block, by shortening it. If alignment hurts, we report negative transfer. If time runs short, optional baselines go first; the core comparison (cube-only policy, unadapted IDM, our method, reference) stays.

Role of group members

All members collaborate and review the report.

Member	Role	Tasks
MEMBER 1	Simulation and data	Block model, expert grasps, data generation, splits
MEMBER 2	IDM and adaptation	IDM, adaptation method, pseudo-label errors
MEMBER 3	Policy and alternatives	Policy, controls, alternative methods
MEMBER 4	Evaluation and integration	Rollouts, statistics, requirement table, figures

Generative-AI disclosure. Codex (OpenAI) and Claude Code (Anthropic) assisted with literature search, bibliography formatting, document structure, and prose. We used them to survey recent work and to format the document; the team revised all text and verified every claim and citation against the original sources.

References

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